

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years building production-grade Node.js and TypeScript services for US-based companies. Specialist in zero-downtime cloud migrations, real-time data synchronization, and RESTful/RPC API design. Led end-to-end delivery of complex integration projects including third-party system migrations and scalable backend services. Experience integrating AI/ML APIs (OpenAI, Google Vision) into production applications. Strong track record with GCP, Firebase, and microservices architecture using Kafka and Pub/Sub.

CORE COMPETENCIES

Node.js & TypeScript

RESTful & RPC APIs

Data Synchronization

Third-Party Integrations

AI/ML API Integration

GCP & Firebase

PostgreSQL & MongoDB

Microservices Architecture

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer · US Remote

- **Led end-to-end migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to CTO
- Designed and built **real-time data synchronization** pipeline between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Built **scalable RESTful APIs** and backend services supporting customer-facing and internal applications
- Implemented Express Shipping feature generating **\$40 additional revenue per order**
- Reduced infrastructure costs by **\$5,000/month** through strategic AWS-to-GCP migration
- Created 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion

MTailor

March 2022 – July 2022

Contractor · US Remote

- Built **third-party integration** scripts to migrate data from MongoDB to Firestore with complex processing logic
- Wrote technical documentation for Node.js and Python Firebase SDKs, enabling team adoption
- Trained Operations team on new Firebase dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer · Singapore Remote

- Built ad-hoc jobs to query **2+ million Firestore records** for real-time stock price updates from Morningstar APIs
- Extended backend APIs using **Firebase Cloud Functions and Express.js** hosted on Heroku
- Developed responsive UI pages for PWA using React and Tailwind CSS

Mind2matter.co

January 2022 – July 2022

Software Engineer · US Remote

- Built **backend services for DeFi applications** using Web3 and Node.js under tight deadlines
- Delivered React UIs from Figma designs for demanding agency clients

Software Engineer · Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected **microservices backend** using Apache Kafka, Docker, and Kubernetes for case management
- Built **gRPC-based USSD service** for legal aid providers communicating with Node.js backend
- Developed cross-platform mobile app using React Native and Expo with Firebase push notifications

PROJECTS

Healthcare WhatsApp Chatbot AI/ML Integration

Integrated OpenAI LLM APIs and Google Vision ML services for prescription OCR via WhatsApp Business API

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Real-time Data Sync Pipeline Data Synchronization

Built live MongoDB ↔ Firestore synchronization supporting parallel production traffic during zero-downtime migration

Node.js · Google Pub/Sub · MongoDB · Firestore

Marketplace Recommendation Engine Backend Services

Built collaborative + content-based filtering engine in MongoDB for service provider matching

Node.js · MongoDB · Express.js

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Backend: Node.js, TypeScript, Express.js, Fastify, Python

Databases: PostgreSQL, MySQL, MongoDB, Firestore, SQL

Cloud: GCP, Firebase, AWS (S3, EC2, EBS), Google Pub/Sub

Infrastructure: Docker, Kubernetes, Apache Kafka, gRPC **Frontend:** React, React Native, Tailwind CSS, HTML/CSS

Other: Git, Linux, Web3, ffmpeg